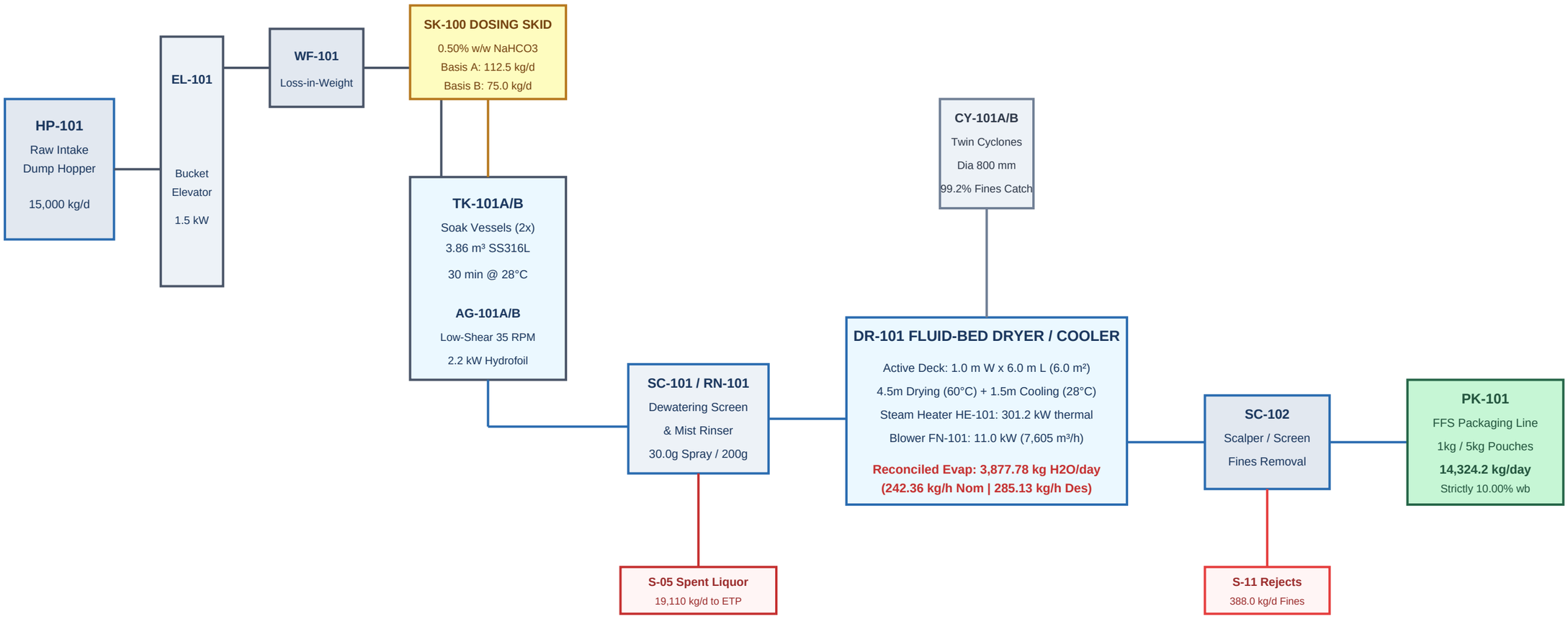
DWG-FIN-15T-002: PROCESS FLOW DIAGRAM & 12-STREAM MASS BALANCE OVERLAY

Project: Harakutir Toor Dal 15 MT/Day Processing Plant | Discipline: Process Engineering | Basis: 15,000 kg/day Incoming Feed



RECONCILED 12-STREAM MASS BALANCE DATA TABLE (Daily Basis: 15,000 kg/day Feed Dal)

Stream ID & Description	Total Mass	Dry Solids	Water	Moisture %	Chemical	Physical State
S-01: Raw Feed Dal (ex-Hopper)	15,000.0 kg	13,425.0 kg	1,575.0 kg	10.50% wb	0.0 kg	Granular Solid
S-02: Fresh Soft Water to Soak	22,387.5 kg	0.0 kg	22,387.5 kg	100.0% wb	0.0 kg	Liquid Water
S-03: Active Chemical (NaHCO3)	112.5 kg	112.5 kg	0.0 kg	0.00% wb	112.5 kg	FCC Powder
S-04: Drained Wet Dal ex-Soak	18,390.0 kg	13,279.0 kg	5,111.0 kg	27.79% wb	23.4 kg	Wet Granules
S-05: Spent Soak Liquor to ETP	19,110.0 kg	258.5 kg	18,851.5 kg	98.65% wb	89.1 kg	Alkaline Effluent
S-06: Fresh Mist Rinse Water	2,250.0 kg	0.0 kg	2,250.0 kg	100.0% wb	0.0 kg	Potable Spray
S-07: Spent Mist Rinse Runoff	2,050.0 kg	38.0 kg	2,012.0 kg	98.15% wb	15.3 kg	Rinse Effluent
S-08: Wet Dal Feed to Dryer	18,590.0 kg	13,241.0 kg	5,349.0 kg	28.77% wb	8.1 kg	Wet Grain Bed
S-09: Hot Fluidizing Air Inlet	129,000.0 kg	127,452.0 kg	1,548.0 kg	1.20% wb	0.0 kg	Air @ 60°C
S-10: Humid Exhaust & Vapor	132,877.8 kg	127,452.0 kg	5,425.8 kg	4.08% wb	0.0 kg	Vapor Exhaust
S-11: Scalped Fines / Rejects	388.0 kg	349.2 kg	38.8 kg	10.00% wb	0.2 kg	Dry Rejects
S-12: Finished Packaged Dal	14,324.2 kg	12,891.8 kg	1,432.4 kg	10.00% wb	7.9 kg	Saleable Product

WARNING: PRELIMINARY — NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT

Design maturity: PRELIMINARY_CALCULATED. Strictly held at Gate G-A pending target lot physical lab data.

HARAKUTIR FOODS LTD — PROCESS ENGINEERING

PROCESS FLOW DIAGRAM & 12-STREAM MASS BALANCE OV

DWG: DWG-FIN-15T-002 | REV: 1.0	SHEET: 1 OF 7
DISC: Process Engineering | UNITS: mm / kg	DATE: 2026-10-15

DWG-FIN-15T-001: PLANT MASTER PLOT PLAN & 5-TIER HYGIENIC ZONING

Project: Harakutir Toor Dal 15 MT/Day Processing Plant | Discipline: Civil / Plant Layout | Basis: 15,000 kg/day Incoming Feed

ZONE 1: RAW INTAKE & STORAGE

Unclassified Hygiene Area

Incoming Grain Storage: 150 MT

Truck Unloading Bay (Roll-up Door)

Intake Hopper HP-101 & Destoner

ZONE 2: WET SOAKING & DOSING

Low Care / Washdown Zone (IP66)

Epoxy Floor, Stainless Trench Drains

Soak Tanks TK-101A/B (Dia 1.6m)

Chemical Dosing Skid SK-100

Dewatering Vibratory Screen SC-101

ZONE 3: THERMAL DRYING

Medium Care Processing Area

Continuous Fluid-Bed Dryer DR-101

Active Deck 1.0m x 6.0m

Blower Fan FN-101 & Steam Coils

Twin Cyclones CY-101A/B

ZONE 4: PACKAGING CLEANROOM

High Care Cleanroom (ISO Class 8)

Positive Pressure (+15 Pa), HEPA Filtered

Personnel Airlock & Gowning Room

Destoner & Scalping Screen SC-102

FFS Packaging Machine PK-101

MCC / ELECTRICAL ROOM

Main 415V Switchboard

PLC & Automation Panel

QA/QC ANALYTICAL LAB

Moisture Analyzers & Flame Photometer

Texture & Cooking Time Test Station

ZONE 5: FINISHED GOODS WAREHOUSE & DISPATCH

Pallet Storage (100 MT Capacity) | Dispatch Roll-Up Door

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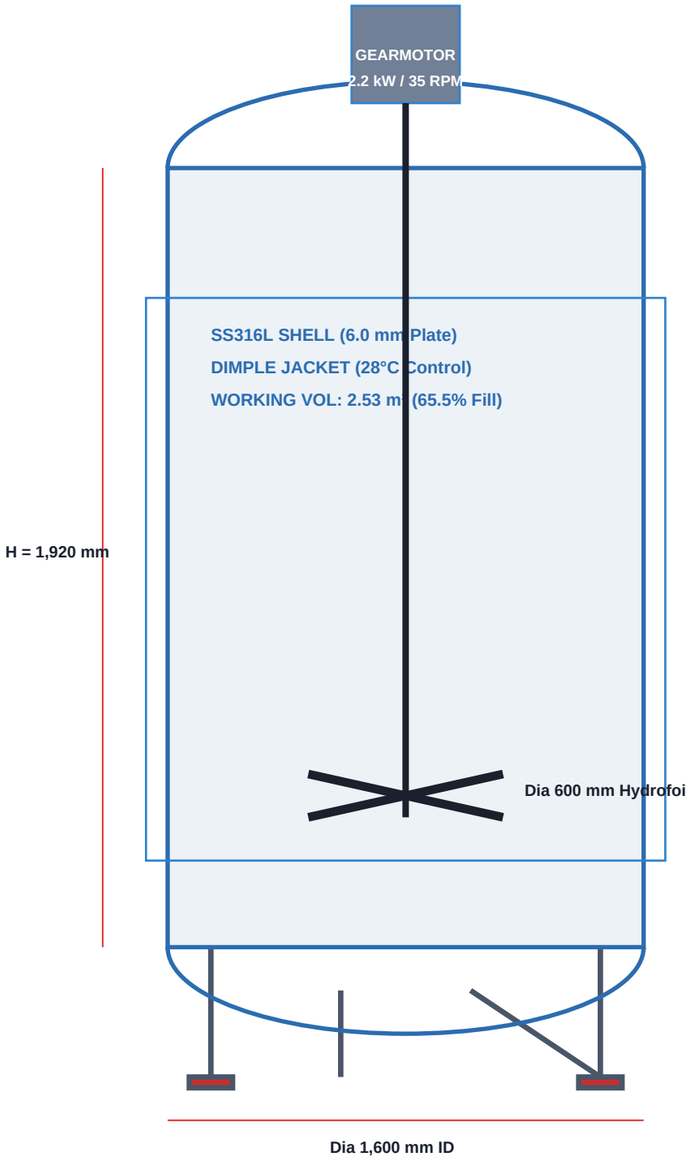
Design maturity: PRELIMINARY_CALCULATED. Strictly held at Gate G-A pending target lot physical lab data.

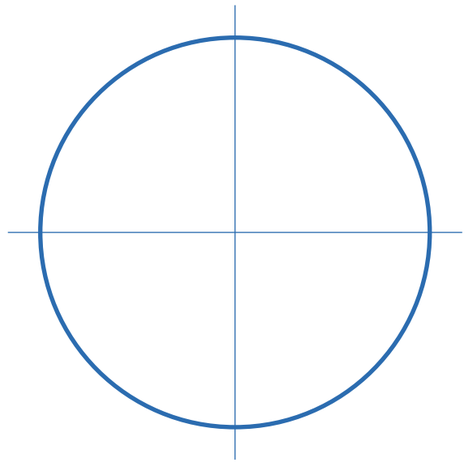
HARAKUTIR FOODS LTD — PROCESS ENGINEERING	
PLANT MASTER PLOT PLAN & 5-TIER HYGIENIC ZONING	
DWG: DWG-FIN-15T-001 | REV: 1.0	SHEET: 2 OF 7
DISC: Civil / Plant Layout | UNITS: mm / kg	DATE: 2026-10-15

DWG-FIN-15T-005: SOAK TANK TK-101A/B MECHANICAL GENERAL ARRANGEMENT

Project: Harakutir Toor Dal 15 MT/Day Processing Plant | Discipline: Mechanical Equipment | Basis: 15,000 kg/day Incoming Feed



PLAN VIEW & NOZZLE ORIENTATIONS



VESSEL NOZZLE SCHEDULE (ASME B16.5 Class 150 RF Flanges)

Tag	Size	Rating	Service Description
N1	DN 250	150# RF	Dal Charge Chute / Manway
N2	DN 50	150# RF	Prepared Chemical Liquor Inlet
N3	DN 80	150# RF	Tank Vent & Breather
N4	DN 65	150# RF	Overflow Nozzle to Sump
N5	DN 150	150# RF	Bottom Slurry Discharge
N6	DN 40	150# RF	CIP Rotary Spray Ball Inlet
N7	DN 25	150# RF	Temperature Sensor Thermowell (RTD)
N8	DN 50	150# RF	Non-Contact Radar Level Transmitter

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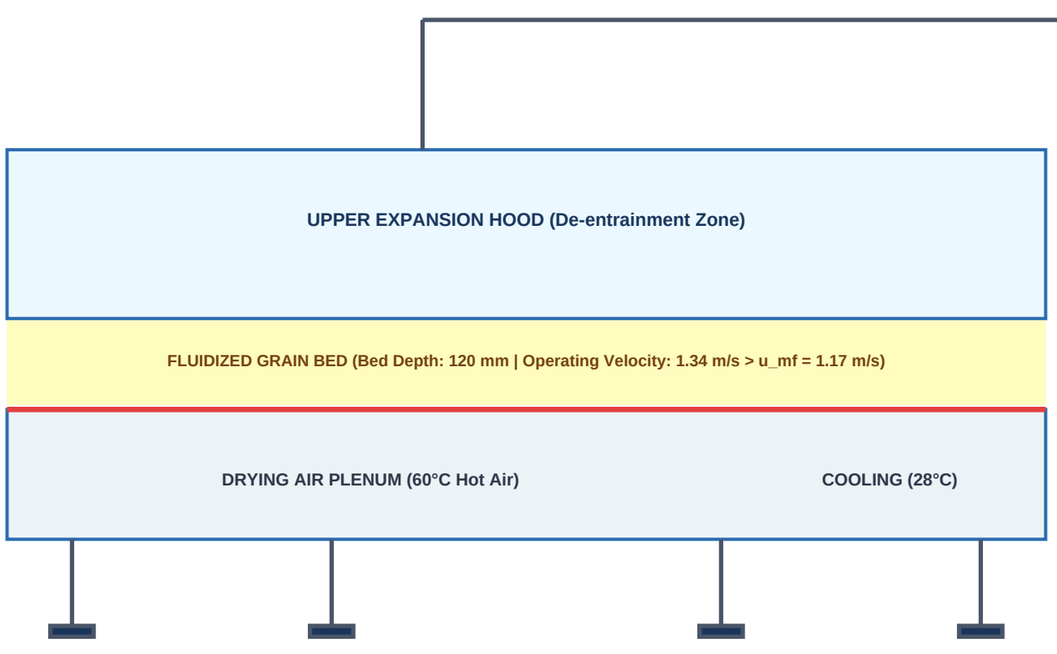
Design maturity: PRELIMINARY_CALCULATED. Strictly held at Gate G-A pending target lot physical lab data.

HARAKUTIR FOODS LTD — PROCESS ENGINEERING	
SOAK TANK TK-101A/B MECHANICAL GENERAL ARRANGEME	
DWG: DWG-FIN-15T-005 | REV: 1.0	SHEET: 3 OF 7
DISC: Mechanical Equipment | UNITS: mm / kg	DATE: 2026-10-15

DWG-FIN-15T-007: CONTINUOUS FLUID-BED DRYER DR-101 GENERAL ARRANGEMENT

Project: Harakutir Toor Dal 15 MT/Day Processing Plant | Discipline: Mechanical Equipment | Basis: 15,000 kg/day Incoming Feed



Total Deck Length = 6,000 mm (4.5m Drying + 1.5m Cooling)

RECONCILED PSYCHROMETRIC & THERMAL DRYING SIZING DATA

Parameter	Design Specification	Reconciled Status
Wet Dal Feed Rate	1,161.9 kg/h (18,590 kg/day @ 16h)	Reconciled from Stream S-08
Inlet Dal Moisture	28.77% wet-basis (5,349 kg/d H2O)	Calculated after Mist Rinse
Exit Dal Moisture	10.00% wet-basis (1,471 kg/d H2O)	Strictly Reconciled Target
Water Evaporation Rate (Nominal)	242.36 kg/h (3,877.78 kg/day @ 16h)	True Rate for 10.00% wb exit
Water Evaporation Rate (Design)	285.13 kg/h (over 13.6h @ 85% OEE)	Equipment Sizing Baseline
Historical Evaporation Rate	230.92 kg/h (under-reported 3,694.7 kg/d)	Defect: Left 183 kg/d water (11.1% wb)
Minimum Fluidization Velocity (u_mf)	1.17 m/s (Wen & Yu Correlation)	Calculated (dp=3.5mm, rho=1350 kg/m³)
Operating Fluidization Velocity (u_0)	1.34 m/s (15% above u_mf + vibration)	Non-channelling gentle bubbling bed
Volumetric Airflow Required	7,605 m³/h (126.8 m³/min)	Air mass flow = 8,060 kg/h @ 60°C
Fluidizing Blower Fan Motor	11.0 kW (Static Head = 2,343 Pa)	Selected Fan Motor Rating
Gross Thermal Heating Duty	301.2 kW thermal (Steam Heat Battery)	Accounts for sensible + latent + 15% loss
Steam Consumption (2.5 bar saturated)	497.2 kg/h saturated steam	Supplied by biomass briquette boiler
Biomass Briquette Fuel Consumption	1.15 MT / day (at 16.5 MJ/kg, 78% eff)	Thermal Fuel OpEx = ■0.522 / kg dal

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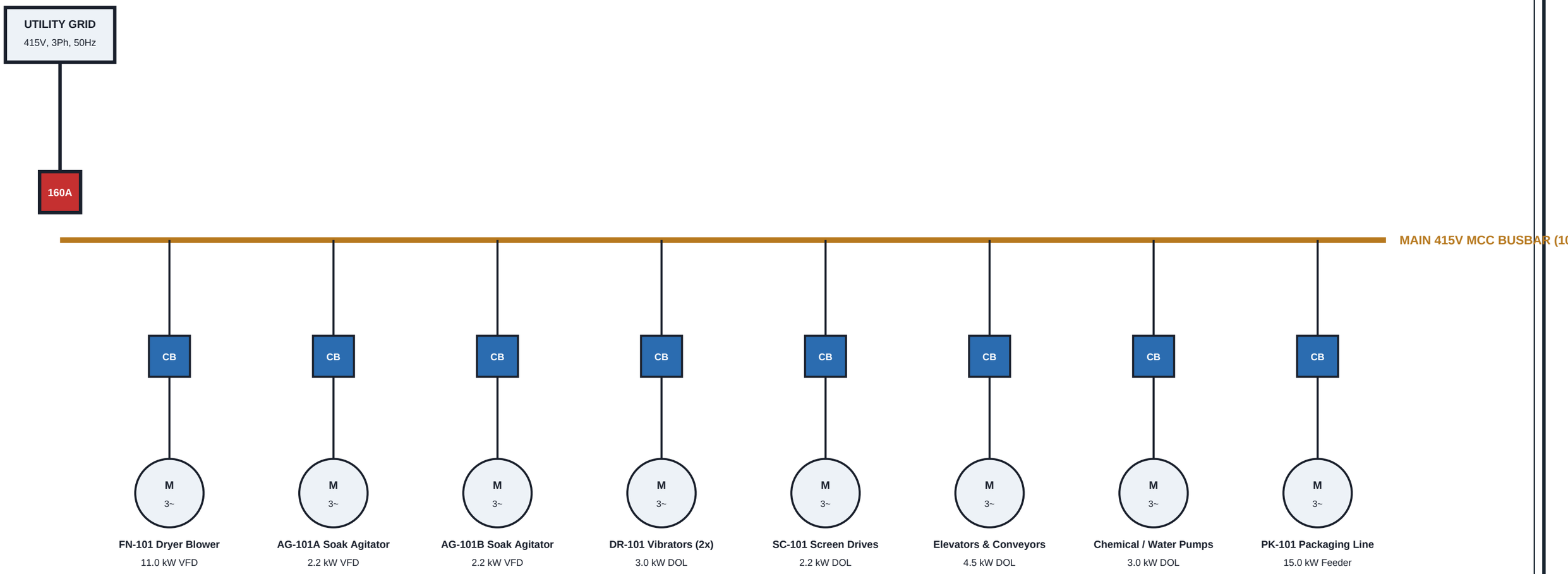
Design maturity: PRELIMINARY_CALCULATED. Strictly held at Gate G-A pending target lot physical lab data.

HARAKUTIR FOODS LTD — PROCESS ENGINEERING	
CONTINUOUS FLUID-BED DRYER DR-101 GENERAL ARRANG	
DWG: DWG-FIN-15T-007 | REV: 1.0	SHEET: 4 OF 7
DISC: Mechanical Equipment | UNITS: mm / kg	DATE: 2026-10-15

DWG-FIN-15T-009: ELECTRICAL SINGLE LINE DIAGRAM & MOTOR CONTROL CENTER

Project: Harakutir Toor Dal 15 MT/Day Processing Plant | Discipline: Electrical Engineering | Basis: 15,000 kg/day Incoming Feed



ELECTRICAL LOAD SUMMARY & DISTRIBUTION

Total Connected Load: 74.69 kW across 31 motors (Master Load List)

Running Load: 50.35 kW (Diversity Factor = 0.674)

Daily Electric Energy: 805.6 kWh/day (16h operating schedule)

Power Factor Correction: APFC Panel included to maintain cos phi >= 0.98

Standby Power: 100 kVA Silent Diesel Generator Set

AUTOMATION, PLC & SAFETY HARDWARE

Controller: Siemens S7-1200 / Allen-Bradley CompactLogix PLC Rack

I/O Capacity: 42 Channels (14 DI, 10 DO, 12 AI, 6 AO per Master IO List)

Operator Interface: 15-inch IP65 Colour Touchscreen HMI Panel

Safety Circuits: Hardwired Dual-Channel Category 3 Emergency Stop Loops

Trips: High-temp shutoff on steam heater, low-pressure blower interlock

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Design maturity: PRELIMINARY_CALCULATED. Strictly held at Gate G-A pending target lot physical lab data.

HARAKUTIR FOODS LTD — PROCESS ENGINEERING	
ELECTRICAL SINGLE LINE DIAGRAM & MOTOR CONTROL C	
DWG: DWG-FIN-15T-009 | REV: 1.0	SHEET: 5 OF 7
DISC: Electrical Engineering | UNITS: mm / kg	DATE: 2026-10-15

REP-NLA-2026-CHART: RECONCILED MASS, MOISTURE & CHEMICAL FORMULATION PROOF

Project: Harakutir Toor Dal 15 MT/Day Processing Plant | Discipline: Process / Quality | Basis: 15,000 kg/day Incoming Feed

1. THE DRYING MOISTURE CONTRADICTION & STEP-BY-STEP MATHEMATICAL PROOF

HISTORICAL REPORT CONTRADICTION (DEFECT IDENTIFIED):

- Reported Total Dal Ex-Dryer: 14,895.30 kg/day
- Reported Water Content: 1,654.30 kg/day (Solids = 13,241.00 kg/day)
- Reported Claimed Moisture: '10.00% wet-basis'
- ARITHMETIC PROOF OF ERROR: 1,654.30 / 14,895.30 = 11.106% wb!
- Result: Left 183.08 kg/d un-evaporated water; under-reported daily evaporation.

RECONCILED MATHEMATICAL TRUTH (AT EXACT 10.00% wb MOISTURE):

- True Total Dal Ex-Dryer: 13,241.00 / (1 - 0.10) = 14,712.22 kg/day
- True Water Content: 14,712.22 x 0.10 = 1,471.22 kg/day
- Verification Check: 1,471.22 / 14,712.22 = EXACTLY 10.0000% wb
- TRUE DAILY EVAPORATION: 5,349.00 - 1,471.22 = 3,877.78 kg H2O/day
- Nominal 16h Rate: 242.36 kg/h | Design 85% OEE Rate (13.6h): 285.13 kg/h

2. CHEMICAL DOSING FORMULATION BASES & ANNUAL OPEX AMBIGUITY

BASIS A: CONCENTRATION IN SOAK LIQUOR (0.50% w/w in Solution)

- Soak Liquor: 15,000 kg feed x 1.50 ratio = 22,500 kg liquor/day
- Active NaHCO3 Required: 22,500 x 0.0050 = 112.50 kg/day
- Concentration on Feed Dal: 112.50 / 15,000 = 0.750% of feed
- Daily Chemical OpEx (@ ■37/kg): ■4,162.50 / day
- Annual Chemical OpEx (300 days): ■12.49 Lakhs / year
- Sizing Rule: Carried as conservative sizing ceiling for Dosing Skid SK-100.

BASIS B: PERCENTAGE OF INCOMING FEED DAL (0.50% w/w on Grain)

- Feed Dal Mass: 15,000 kg dal/day
- Active NaHCO3 Required: 15,000 x 0.0050 = 75.00 kg/day
- Concentration in Soak Liquor: 75.00 / 22,500 = 0.333% in solution
- Daily Chemical OpEx (@ ■37/kg): ■2,775.00 / day
- Annual Chemical OpEx (300 days): ■8.33 Lakhs / year
- Variance: 37.50 kg/day difference (■4.16 Lakhs/year OpEx swing).

3. WHOLE-PLANT DRY SOLIDS RECOVERY & CONTRADICTION RECONCILIATION

- 15,000 kg/day Feed Dal at 10.5% moisture contains exactly 13,425.00 kg/day Dry Solids.
- Soak Leaching Loss: 1.46% of feed solids = 196.01 kg/day dissolved solids lost in soak liquor (98.54% wash leaching recovery).
- Mechanical Screening / Scalping Loss: 2.89% of feed solids = 387.98 kg/day cracked cotyledon fines collected by aspirators.
- Total Plant Dry Solids Lost: 196.01 + 387.98 = 583.99 kg/day (4.35% total dry solids loss across factory).
- Net Whole-Plant Dry Solids Recovery: 13,425.00 - 583.99 = 12,841.01 kg/day (EXACTLY 95.65% WHOLE-PLANT RECOVERY).
- Final Saleable Finished Product: 12,841.01 / 0.90 = 14,267.79 kg/day at strictly 10.00% wet-basis moisture.

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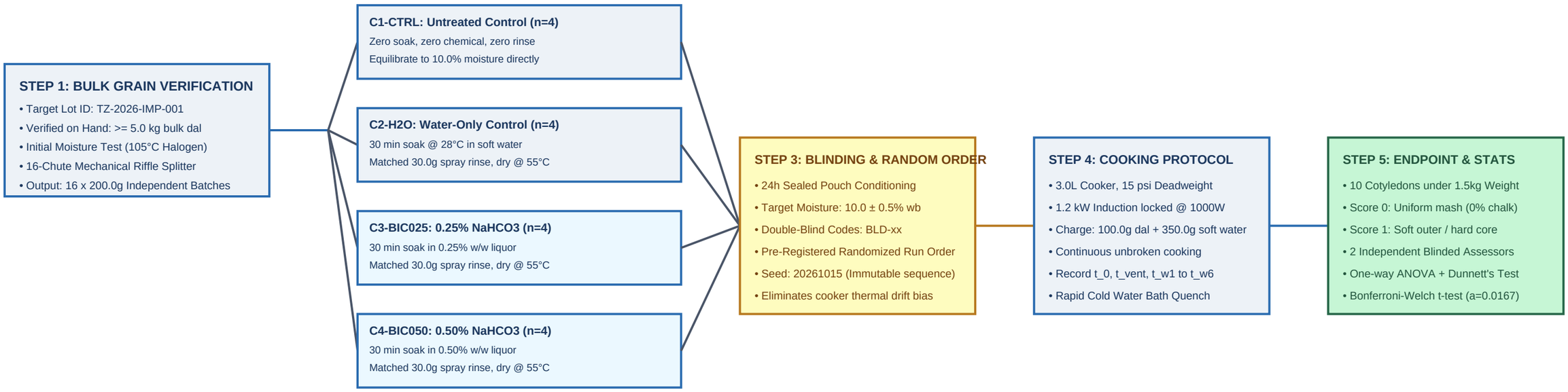
HARAKUTIR FOODS LTD — PROCESS ENGINEERING	
RECONCILED MASS, MOISTURE & CHEMICAL FORMULATION	
DWG: REP-NLA-2026-CHART | REV: 1.0	SHEET: 6 OF 7
DISC: Process / Quality | UNITS: mm / kg	DATE: 2026-10-15

PROTOCOL-SCR-2026: PRE-REGISTERED 4-CONDITION BENCH SCREENING PROTOCOL

Project: Harakutir Toor Dal 15 MT/Day Processing Plant | Discipline: Research / Quality | Basis: 15,000 kg/day Incoming Feed



PRE-REGISTERED DECISION RULES, ACCEPTANCE GATES & REGULATORY STOP CRITERIA

Criterion	Target Lot Acceptance Boundary	Immediate Disqualification / Action
Cooking Time to Score 0	<= 15.0 minutes (<= 3.5 whistles)	FAIL if > 16.5 minutes (Triggers non-chemical milling/steam route)
Sensory Flavor Acceptance	>= 6.0 / 9.0 on Hedonic Scale	DISQUALIFIED if any soapy/chemical off-flavor detected (Score < 5.0)
Whole-Plant Solids Recovery	>= 95.50% (Leaching loss <= 3.0%)	FAIL if leaching loss > 5.0% (Exceeds economic viability bound)
Retained Elemental Sodium	< 120.0 mg Na / 100g dal	NON-COMPLIANT if >= 350.0 mg Na / 100g (Violates FSSAI low sodium standard)
Statistical Significance	t_obs >= 2.998 (p <= 0.0167 vs Control)	No commercial claim permitted if effect is statistically indistinguishable
FSSAI Labeling Mandate	Declare: 'Acidity Regulator (INS 500(ii))'	ILLEGAL to treat as undeclared processing aid; label sign-off mandatory

GATE G-A TRIAL COMMENCEMENT AUTHORIZATION (HOLD POINT):

Status: PROTOCOL_READY_FOR_REVIEW. The protocol is frozen and data sheets are blank.

Laboratory personnel are strictly prohibited from entering simulated data or fabricating observations.

Execution awaits physical sample acquisition of 5.0 kg lot TZ-2026-IMP-001 and written human lead authorization.

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Design maturity: PRELIMINARY_CALCULATED. Strictly held at Gate G-A pending target lot physical lab data.

HARAKUTIR FOODS LTD — PROCESS ENGINEERING	
PRE-REGISTERED 4-CONDITION BENCH SCREENING PROTO	
DWG: PROTOCOL-SCR-2026 | REV: 1.0	SHEET: 7 OF 7
DISC: Research / Quality | UNITS: mm / kg	DATE: 2026-10-15